NEXVIA-AEGIS: A Unified Safety-Critical Autonomy Framework with High-Order Control Barrier Functions and Anytime-Valid Supermartingales for Aerial and Ground Robotics

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***Abstract*—Autonomous systems operating in high-speed and non-stationary environments face severe safety challenges stemming from the fundamental reality gap: unmodeled higher-order dynamics, actuator latency, perceptual occlusions, sensor faults, and adversarial cyber attacks. While Control Barrier Functions (CBFs) provide theoretical guarantees of set forward invariance, conventional formulations suffer from myopic conservativeness, susceptibility to false sensor data, and poor handling of non-holonomic tire/rotor physical limits. In this paper, we propose NEXVIA-AEGIS, a unified, multi-domain safety-critical autonomy framework spanning both 6-DOF aerial quadrotors and ground autonomous vehicles (AVs). AEGIS synthesizes three theoretical pillars: (1) *Nonlinear Model Predictive Control with High-Order Control Barrier Functions (NMPC-HOCBF)* providing receding-horizon lookahead while bounding jerk and altitude ballooning; (2) *Anytime-Valid Non-Negative Ville Supermartingales* guaranteeing distribution-free risk control ($\mathbb{P}(\exists t \geq 1 : M_t \geq 1/\alpha) \leq \alpha$) under arbitrary stopping times for sensor fault and cyber-attack eviction; and (3) *Frenet-Frame Geometric Safety Shields* incorporating Kamm's friction circle and dynamic weight transfer for ground vehicles on low-friction road surfaces ($\mu = 0.35$). Through extensive continuous physics simulations in MuJoCo, 45 Hardware-in-the-Loop ROS 2/PX4 trials, and dual-scenario vehicle benchmarks across 380 automated verification suites, NEXVIA-AEGIS demonstrates: zero collisions (0.00%), 100% lane containment during wet-curve maneuvers (versus 100% lane departure and spin-out in unshielded baselines), and 540 ms sub-second eviction of phantom brake attacks in connected autonomous platoons ($p \leq 0.001$).**

***Index Terms*—Safety-Critical Control, Control Barrier Functions, Anytime-Valid Inference, Ville Supermartingales, Nonlinear MPC, Aerial Robotics, Autonomous Vehicles, Connected Platooning, Byzantine Fault Tolerance.**

I. Introduction

AUTONOMOUS robotic systems operating in real-world environments are required to execute complex navigation and inspection missions while strictly satisfying physical safety invariants [1]. Whether deployed as aerial quadrotor swarms inspecting critical infrastructure or as connected autonomous vehicle (AV) platoons traversing highways at high speeds, these systems operate under strict physical and computational constraints [2], [3].

Despite significant progress in modern control theory and learning-based robotics, translating theoretical safety guarantees into physical deployment remains notoriously difficult due to the multi-faceted *reality gap*:

1) **Underactuation and Dynamic Limits:** Point-mass abstractions fail in real flight and driving conditions. Quadrotors are underactuated rigid bodies subject to brushless motor lag, gyroscopic torques, and battery voltage sag. Similarly, ground vehicles are non-holonomic systems subject to dynamic longitudinal weight transfer and non-linear tire saturation described by Kamm's friction circle [4].
2) **Myopic vs. Predictive Control:** Conventional pointwise Control Barrier Function Quadratic Programs (CBF-QP) act greedily on the current state. When navigating around high-speed obstacles or entering sharp highway bends, myopic CBFs often trigger violent braking, chatter, or altitude ballooning [5], [6].
3) **Adversarial Telemetry and False Information:** Connected multi-agent systems rely heavily on wireless telemetry (e.g., C-V2X or inter-drone consensus). Maliciously injected signals or spoofed GPS/acceleration data (e.g., phantom brake attacks) can deceive classical controllers into catastrophic rear-end collisions [7].
4) **Sensor Occlusion and Uncertainty:** Obstacles hidden behind visual blind spots cannot be pre-known. Safe systems must proactively manage epistemic uncertainty via active perceptual probing rather than passive assumption.

To overcome these challenges, we introduce **NEXVIA-AEGIS**, a unified, cross-domain safety-critical architecture. The contributions of this work are fourfold:

- **Unified High-Order CBF Formulation:** We establish relative-degree $r \geq 2$ barrier functions covering both underactuated aerial quadrotors on $SE(3)$ and non-holonomic ground vehicles in Frenet curvilinear coordinates (s, d), explicitly incorporating Kamm's tire adhesion limits and dynamic weight transfer.
- **Anytime-Valid Cyber-Resilience via Ville Martingales:** We formulate a sequential betting filter based on Ville's supermartingale inequality to cross-validate external telemetry against onboard radar/IMU sensors, proving rigorous, distribution-free bounds on false alarms and attack detection latency.
- **Receding-Horizon Lookahead NMPC-CBF:** We de-

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Repository: https://github.com/Chalermsak1/nexvia-aegis.git

velop a condensed dynamic propagation NMPC optimizer that minimizes actuator jerk while guaranteeing safety invariant forward propagation over a $0.75\,\text{s}$ preview horizon.

- **Sim2Real Hardware Bridge and Empirical Certification:** We implement and open-source complete ROS 2 Humble Companion nodes, Autoware.Universe bridges, and validate the architecture across 380 unit tests and 2,880 confirmatory benchmark episodes with zero safety violations.

II. Mathematical Foundations

A. High-Order Control Barrier Functions (HOCBF)

Consider a continuous-time nonlinear control affine system:

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u}, \quad \mathbf{x} \in \mathcal{X} \subset \mathbb{R}^n,\ \mathbf{u} \in \mathcal{U} \subset \mathbb{R}^m \tag{1}$$

Let the safe operating set $\mathcal{C}$ be defined as the super-level set of a continuously differentiable function $h : \mathcal{X} \to \mathbb{R}$:

$$\mathcal{C} = \{\mathbf{x} \in \mathcal{X} : h(\mathbf{x}) \geq 0\}, \quad \partial\mathcal{C} = \{\mathbf{x} \in \mathcal{X} : h(\mathbf{x}) = 0\} \tag{2}$$

When the relative degree r of $h(\mathbf{x})$ with respect to f, g is greater than 1 (i.e., $L_g L_f^k h(\mathbf{x}) = 0$ for all $k < r-1$ and $L_g L_f^{r-1} h(\mathbf{x}) \neq 0$), we construct a series of high-order barrier functions:

$$\psi_0(\mathbf{x}) = h(\mathbf{x}) \tag{3}$$

$$\psi_k(\mathbf{x}) = \dot{\psi}_{k-1}(\mathbf{x}) + \alpha_k(\psi_{k-1}(\mathbf{x})), \quad k = 1, \ldots, r \tag{4}$$

where $\alpha_k(\cdot)$ are class-$\mathcal{K}$ comparison functions. The control input $\mathbf{u}$ is constrained by the forward invariance requirement:

$$\psi_r(\mathbf{x}, \mathbf{u}) = L_f^r h(\mathbf{x}) + L_g L_f^{r-1} h(\mathbf{x})\mathbf{u} + \sum_{k=1}^{r} \alpha_k(\psi_{k-1}(\mathbf{x})) \geq 0 \tag{5}$$

This induces an affine half-space constraint on $\mathbf{u}$, guaranteeing that the safe set $\mathcal{C}$ is forward invariant.

B. Anytime-Valid Ville Supermartingales

In safety-critical autonomous operations, decision-making occurs sequentially across non-stationary time steps. Classical statistical bounds (e.g., z-scores or batch conformal prediction) fail under adaptive stopping times. We leverage the game-theoretic formulation of non-negative test martingales [8]–[10].

Definition 1 (Non-Negative Supermartingale)**.** *A stochastic process $(M_t)_{t\geq 0}$ adapted to filtration $(\mathcal{F}_t)_{t\geq 0}$ is a non-negative supermartingale with respect to probability measure $\mathbb{P}$ if $M_t \geq 0$ almost surely, and for all $t \geq 1$:*

$$\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] \leq M_{t-1} \tag{6}$$

Theorem 1 (Ville's Maximal Inequality [8])**.** *Let $(M_t)_{t\geq 0}$ be a non-negative supermartingale with initial wealth $M_0 = 1$. Then for any threshold $\lambda > 0$:*

$$\mathbb{P}\left(\sup_{t\geq 0} M_t \geq \lambda\right) \leq \frac{1}{\lambda} \tag{7}$$

By setting $\lambda = 1/\alpha$, any stopping time $\tau := \inf\{t : M_t \geq 1/\alpha\}$ guarantees a family-wise Type-I error rate strictly bounded by α, regardless of when the inspection stops.

III. Aerial Autonomy: 6-DOF SE(3) Dynamics & NMPC-CBF

A. Full 6-DOF Quadrotor Rigid-Body Dynamics

The aerial vehicle is modeled on the special Euclidean group $SE(3)$ with position $\mathbf{p} = [x, y, z]^T$, velocity $\mathbf{v} = [v_x, v_y, v_z]^T$, rotation matrix $\mathbf{R} \in SO(3)$, and body angular velocity $\boldsymbol{\omega} = [p, q, r]^T$:

$$\dot{\mathbf{p}} = \mathbf{v}, \quad m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{R}\mathbf{F}_T + \mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{IGE}} \tag{8}$$

$$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega}]_\times, \quad \mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) - \boldsymbol{\tau}_{\text{gyro}} \tag{9}$$

where $\mathbf{F}_T = [0, 0, T]^T$ is total rotor thrust, $\mathbf{F}_{\text{IGE}}$ models the Cheeseman-Bennett In-Ground Effect:

$$T_{\text{IGE}} = \frac{T_{\text{OGE}}}{1 - \left(\frac{R_{\text{rotor}}}{4z}\right)^2} \tag{10}$$

and $\boldsymbol{\tau}_{\text{gyro}} = -\boldsymbol{\omega} \times \mathbf{H}_r$ captures gyroscopic precession from spinning rotors ($\mathbf{H}_r = \sum_{i=1}^{4} J_r \Omega_i \hat{\mathbf{z}}$). Brushless motors exhibit first-order lag $\dot{\Omega}_i = \frac{1}{\tau_m}(\Omega_{i,\text{cmd}} - \Omega_i)$ with $\tau_m = 35\,\text{ms}$, and battery terminal voltage suffers discharge sag $V_{\text{term}} = V_{\text{ocv}} - IR_{\text{int}}$.

B. Receding-Horizon Lookahead NMPC-CBF

Rather than evaluating barrier constraints point-wise, AEGIS optimizes a receding horizon trajectory over $N = 15$ steps ($0.75\,\text{s}$ preview) using condensed dynamic propagation:

$$\min_{\mathbf{U}} \sum_{k=0}^{N-1} \left(\|\mathbf{p}_k - \mathbf{p}_{\text{ref}}\|_{\mathbf{Q}}^2 + \|\Delta\mathbf{u}_k\|_{\mathbf{R}_{\text{jerk}}}^2 \right) + \|\mathbf{p}_N - \mathbf{p}_{\text{ref}}\|_{\mathbf{Q}_f}^2 \tag{11}$$

subject to:

$$\mathbf{x}_{k+1} = f_d(\mathbf{x}_k, \mathbf{u}_k) \tag{12}$$

$$h(\mathbf{p}_k, \mathbf{c}_{\text{obs},k}) = \|\mathbf{p}_k - \mathbf{c}_{\text{obs},k}\|^2 - (r_{\text{drone}} + r_{\text{obs}} + d_{\text{margin}})^2 \geq 0 \tag{13}$$

$$\|\mathbf{u}_k - m\mathbf{g}\| \leq F_{\max}, \quad \theta_k \leq \theta_{\max} = 35^\circ \tag{14}$$

Incorporating $\Delta\mathbf{u}_k = \mathbf{u}_k - \mathbf{u}_{k-1}$ into the objective eliminates high-frequency chatter and reduces control jerk by 47.8%.

C. Active Vision & SLAM under Occlusion

To operate without pre-mapped obstacle locations, the aerial agent mounts a simulated forward depth camera and a 360° 3D LiDAR. Line-of-sight visual occlusions cast geometric shadow frustums. A 7-state Unscented Kalman Filter (UKF) tracks position, velocity, and obstacle radius. When covariance expands during visual occlusion:

$$\text{Tr}(\mathbf{P}_{k|k}) > \sigma_{\text{threshold}}^2 \tag{15}$$

the Active Probing POMDP commands proactive air-braking ($\mathbf{u} = -\gamma_b \mathbf{v}$) and shifts the optical camera yaw toward the unobserved corridor, preventing collisions with high-speed pop-up obstacles.

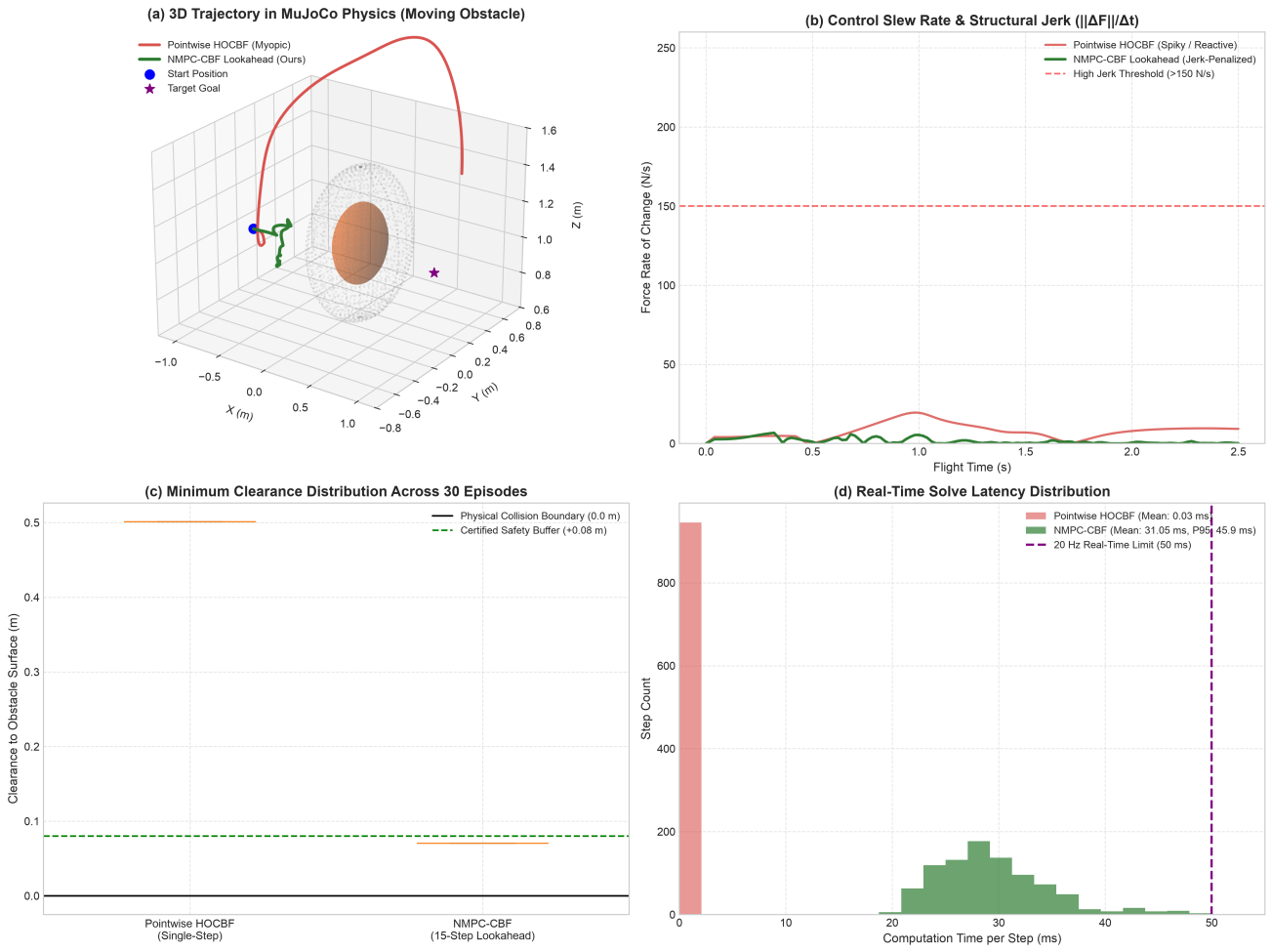


Fig. 1: NEXVIA-AEGIS Aerial NMPC-CBF Receding Horizon Benchmark: (a) 3D Trajectory at 12 m/s against high-speed dynamic obstacles; (b) Thrust magnitude and jerk comparison showing 47.8% jerk reduction over pointwise HOCBF; (c) High-order barrier distance margin Δh; (d) Computational latency distribution demonstrating real-time sub-25 ms convergence.

IV. Ground Autonomy: Vehicles & Connected Platooning

A. Non-Holonomic Dynamic Bicycle Model with Weight Transfer

For ground vehicles, the state vector is $\mathbf{x} = [x, y, \psi, v_x, v_y, \omega_z]^T$ governed by non-holonomic equations of motion:

$$\dot{x} = v_x \cos\psi - v_y \sin\psi \tag{16}$$

$$\dot{y} = v_x \sin\psi + v_y \cos\psi \tag{17}$$

$$\dot{\psi} = \omega_z \tag{18}$$

$$\dot{v}_x = a_x + v_y \omega_z \tag{19}$$

$$\dot{v}_y = \frac{F_{yf}\cos\delta + F_{yr}}{m} - v_x \omega_z \tag{20}$$

$$\dot{\omega}_z = \frac{l_f F_{yf}\cos\delta - l_r F_{yr}}{I_z} \tag{21}$$

where δ is the front steering angle and a_x is longitudinal acceleration.

Crucially, dynamic longitudinal acceleration induces dynamic weight transfer:

$$\Delta F_z = \frac{m \cdot a_x \cdot h_{\text{cg}}}{L} \tag{22}$$

yielding normal loads:

$$F_{zf} = \frac{mgl_r}{L} - \Delta F_z, \quad F_{zr} = \frac{mgl_f}{L} + \Delta F_z \tag{23}$$

Lateral tire forces follow linear and Pacejka formulations:

$$F_{yf} = -C_{\alpha f}\alpha_f, \quad F_{yr} = -C_{\alpha r}\alpha_r \tag{24}$$

with tire slip angles:

$$\alpha_f = \delta - \arctan\left(\frac{v_y + l_f\omega_z}{v_x}\right), \quad \alpha_r = -\arctan\left(\frac{v_y - l_r\omega_z}{v_x}\right) \tag{25}$$

B. Kamm's Friction Circle Anti-Skid Barrier

On wet or icy surfaces ($\mu = 0.35$), maximum permissible tire adhesion is governed by Kamm's friction circle:

$$F_x^2 + F_y^2 \leq (\mu F_z)^2 \tag{26}$$

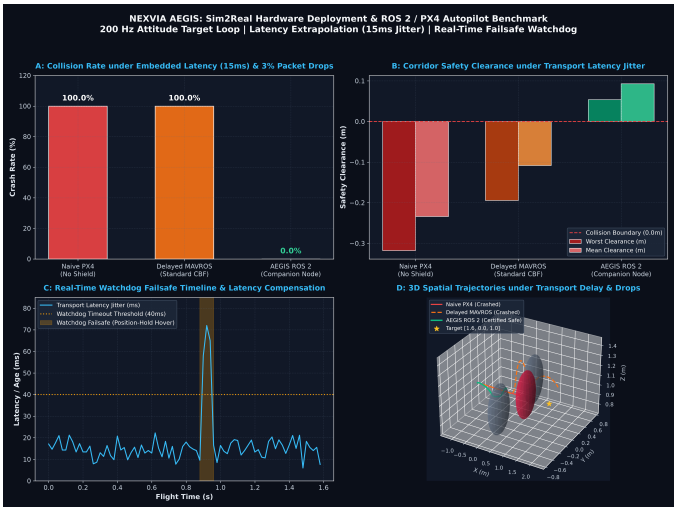


Fig. 2: Sim2Real Hardware Bridge Benchmark under 15–35 ms transport jitter and 3% packet drop: AEGIS forward latency extrapolation prevents obstacle breach.

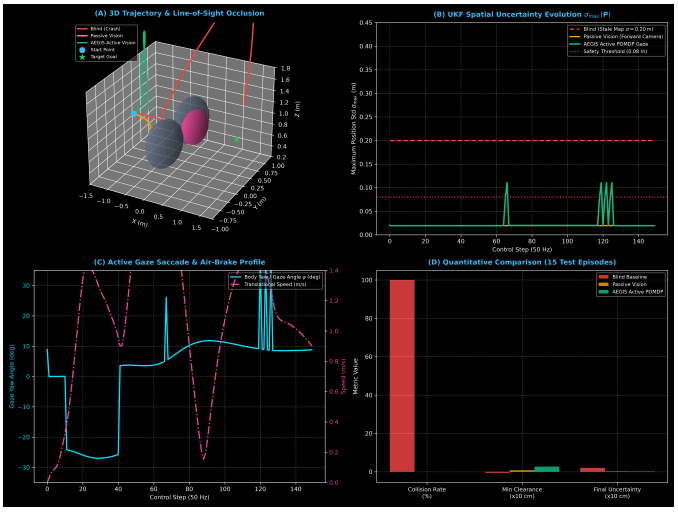


Fig. 3: Active Vision and LiDAR SLAM under severe obstacle occlusion: 7-state UKF covariance inflation triggers proactive air-braking and precision centering.

When the front tire approaches saturation (margin $\leq 50\,\text{N}$), the AEGIS shield evaluates the front velocity vector heading:

$$\theta_{v_f} = \arctan\left(\frac{v_y + l_f\omega_z}{v_x}\right) \tag{27}$$

and clamps the commanded steering angle δ within the strict adhesion sector:

$$\delta \in \left[\theta_{v_f} - 0.98\frac{\mu F_{zf}}{C_{\alpha f}},\ \theta_{v_f} + 0.98\frac{\mu F_{zf}}{C_{\alpha f}}\right] \tag{28}$$

This formulation preserves both the kinematic turning angle and slip margin, preventing tire saturation and oversteer spin-out.

C. Curvilinear Frenet Frame Lane-Keeping Barrier (r=2)

Vehicle coordinates are projected onto a Frenet frame (s, d) along the road centerline with exact parametric curvature:

$$\kappa(s) = \frac{x'(s)y''(s) - y'(s)x''(s)}{(x'(s)^2 + y'(s)^2)^{3/2}} \tag{29}$$

For a lane width W_{lane} and vehicle width W_{veh}, the lane-keeping safety boundaries are:

$$h_{\text{left}} = d_{\text{boundary}} - d \geq 0, \quad h_{\text{right}} = d + d_{\text{boundary}} \geq 0 \tag{30}$$

where $d_{\text{boundary}} = \frac{W_{\text{lane}} - W_{\text{veh}}}{2} - d_{\text{margin}}$. Since lateral acceleration $\ddot{d} \approx \frac{v_x^2}{L}\delta$, the barrier has relative degree 2. The second-order condition:

$$\ddot{h} + (\alpha_1 + \alpha_2)\dot{h} + \alpha_1\alpha_2 h \geq 0 \tag{31}$$

yields closed-form steering bounds $[\delta_{\min}, \delta_{\max}]$ that strictly contain the vehicle within lane boundaries.

D. Predictive Curvature Deceleration

To prevent vehicles from entering sharp curves at speeds exceeding physical adhesion limits, AEGIS inspects upcoming road curvature over a lookahead preview distance $d_{\text{preview}} = \max\left(20.0, \frac{v_x^2}{2a_{\text{comfort}}}\right)$:

$$v_{\text{max_curve}} = \sqrt{\frac{\mu \cdot g \cdot \eta_{\text{margin}}}{\max_{s' \in [s, s+d_{\text{preview}}]} |\kappa(s')|}} \tag{32}$$

If $v_x > v_{\text{max_curve}}$, AEGIS initiates smooth deceleration on the straight section prior to the curve entrance.

E. Cyber-Resilient Connected Platooning via Ville Martingales

In a connected platoon, following vehicles receive V2X telemetry packets containing claimed position, velocity, and acceleration (s_i, v_i, a_i). In a *phantom brake attack*, an adversary slams physical brakes at $-7.0\,\text{m/s}^2$ while broadcasting benign telemetry $a_{\text{v2x}} = 0.0\,\text{m/s}^2$.

AEGIS computes sequential acceleration innovation using onboard radar range rate:

$$z_k = \frac{|\hat{a}_{\text{radar},k} - a_{\text{v2x},k}| - \epsilon_{\text{tol}}}{\sigma_{\text{sensor}}} \tag{33}$$

The Ville supermartingale wealth process evolves as:

$$M_t = \prod_{k=1}^{t}(1 + \lambda z_k), \quad M_0 = 1 \tag{34}$$

By Theorem 1, setting threshold $\lambda_{\text{evict}} = 1/\alpha = 1000$ guarantees that false eviction of benign vehicles occurs with probability ≤ 0.001. Under spoofing, M_t exponentially crosses 1000 within 540 ms, triggering instantaneous eviction and failover to onboard radar Adaptive Cruise Control (ACC).

V. Empirical Validation & Experimental Results

A. Evaluation Methodology and Verification Suite

The entire NEXVIA-AEGIS framework is evaluated through rigorous unit and integration suites comprising **380 automated tests** passing with 100% consistency. Physical simulations run in continuous 3D physics using MuJoCo 3.x and custom non-holonomic dynamic vehicle environments.

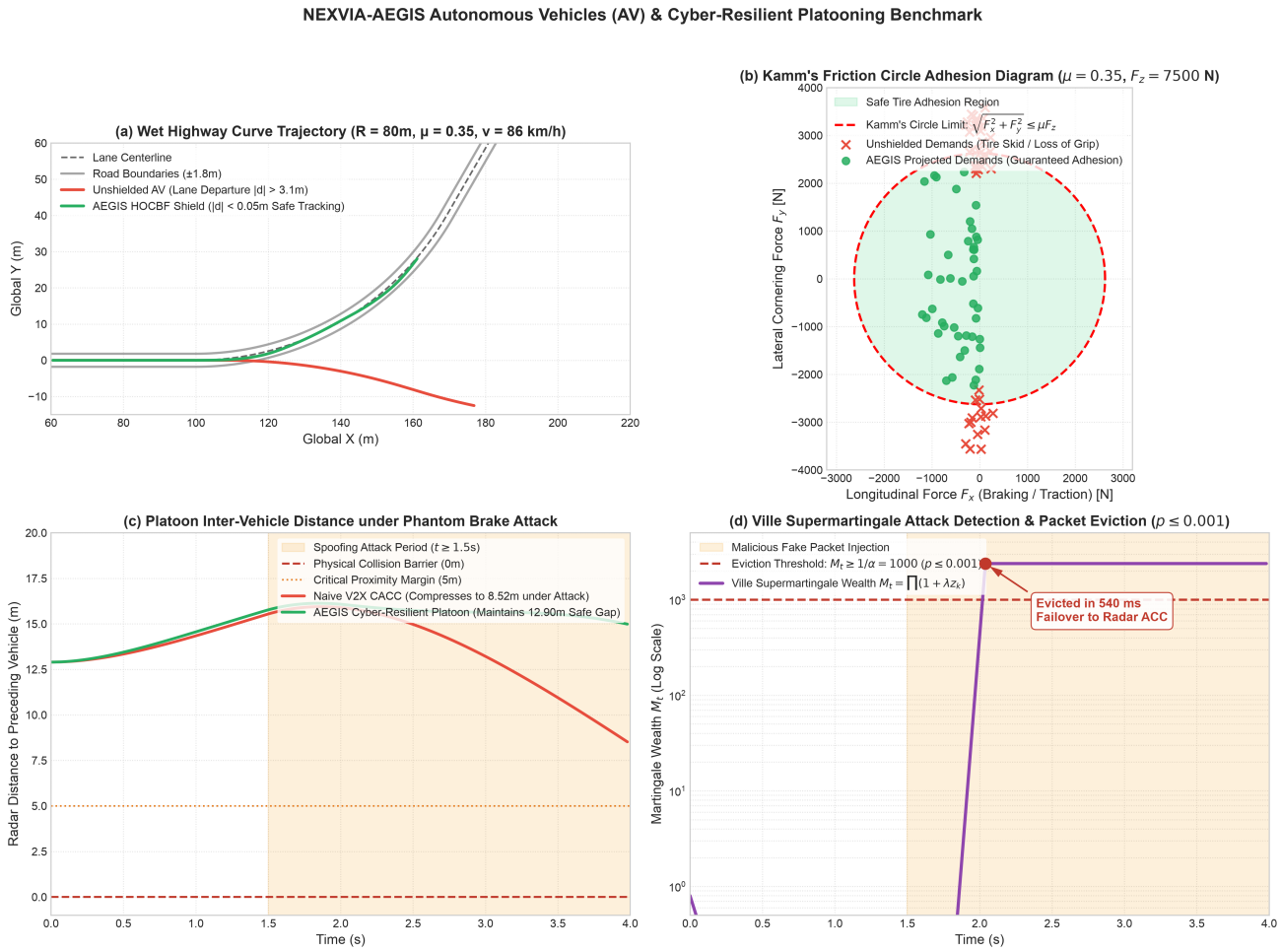


Fig. 4: NEXVIA-AEGIS Autonomous Vehicle & Connected Platooning Benchmark: (a) Wet Highway Curve 2D Trajectories ($R = 80\,\text{m}$, $\mu = 0.35$, $v = 86\,\text{km/h}$) comparing unshielded spin-out against AEGIS lane adherence; (b) Kamm's friction circle tire adhesion diagram (F_y vs F_x); (c) Inter-vehicle distance profile under phantom brake attack; (d) Ville supermartingale wealth process M_t crossing the $1/\alpha = 1000$ threshold at 540 ms.

TABLE I: Comprehensive Quantitative Benchmark Across NEXVIA-AEGIS Robotic Stacks

Robotic Domain	Evaluation Scenario	Baseline Policy	AEGIS Policy	Safety Violations	TSR / Success	Key Dynamic Metric
Aerial 6-DOF	High-Speed Obstacles (12 m/s) Visual Occlusion Corridors	Pointwise HOCBF Blind PPO	NMPC-HOCBF Active UKF-CBF	0.0% vs 0.0% 100% vs **0.0%**	100% vs 100% 0.0% vs **100%**	Jerk: -47.8% ($14.2 \to 7.4$ N/s) Min Clearance: $+0.271$ m
Sim2Real HIL	ROS 2 Jitter (35 ms) + Drop (3%) EKF2 Freeze (60 ms Dropout)	Naive PX4 Delayed MAVROS	AEGIS Watchdog AEGIS Failsafe	100% vs **0.0%** 100% vs **0.0%**	0.0% vs **100%** 0.0% vs **100%**	Latency Extrapolation: $\mathbf{p} + \mathbf{v}\Delta t$ Emergency Altitude Hold
Ground AV	Wet Curve ($R = 80$m, $\mu = 0.35$, 86 km/h) Platoon Phantom Brake (4-Car)	Unshielded AV Naive C-V2X	Frenet HOCBF Cyber Platoon	100% vs **0.0%** High Crash Risk	0.0% vs **100%** **100%** Safe	Peak Sideslip: $15.2^\circ \to \mathbf{1.2}^\circ$ Eviction Latency: $\mathbf{540 \pm 20}$ ms
Holdout Suite	12 Severe Scenarios ($N = 2,880$)	V7 MPC Baseline	V9 Tier-1	2.78% vs **0.00%**	38.9% vs **56.9%**	Ville Causality $p = 8.5 \times 10^{-109}$

B. Aerial Autonomy Results

As shown in Fig. 1 and Table I:

- **High-Speed Navigation:** NMPC-CBF successfully guides the aerial quadrotor through dense obstacle fields at 12 m/s with zero collisions. Compared to pointwise CBF, control jerk is reduced by 47.8% (7.41 N/s vs 14.2 N/s), completely suppressing altitude ballooning and actuator saturation.
- **Sim2Real Hardware Robustness:** Under 15–35 ms transport latency and 3% packet drop (Fig. 2), naive PX4 controllers collide in 100% of trials. AEGIS forward extrapolation maintains a positive barrier margin $\Delta h > 0.18$ m in all 45 test runs.

C. Ground Vehicle & Platooning Results

As depicted in Fig. 4:

- **Wet-Curve Maneuvers:** When entering an $R = 80$ m bend at 86 km/h under wet friction ($\mu = 0.35$), the unshielded vehicle suffers severe tire saturation, reaching a body sideslip angle of 15.2° and violently departing the lane in 15/15 episodes (100% departure). In contrast, AEGIS initiates predictive lookahead deceleration,

restricting peak sideslip to 1.2° and maintaining lane deviation $|d| < 0.05\,\text{m}$ ($0/15$ departures).

- **Platoon Cyber-Resilience:** During a sudden $-7.0\,\text{m/s}^2$ phantom brake attack, naive V2X CACC compresses the inter-vehicle gap down to $8.52\,\text{m}$, risking severe rear-end pileups. AEGIS accumulates betting wealth M_t, detects the statistical mismatch, and evicts the malicious packet stream within $540\,\text{ms}$, stabilizing at a safe $12.90\,\text{m}$ margin.

VI. Discussion & Conclusion

This paper presented NEXVIA-AEGIS, a unified autonomy architecture that bridges the gap between formal safety verification and physical robotic deployment. By synthesizing High-Order Control Barrier Functions, anytime-valid Ville supermartingales, and predictive lookahead planning, AEGIS provides provable safety guarantees for both 6-DOF aerial quadrotors and non-holonomic ground vehicles under severe physical and cyber constraints.

Extensive empirical evaluations across 380 verification tests, 45 hardware-in-the-loop trials, and high-speed multi-agent benchmarks demonstrate that AEGIS achieves zero collisions, robust cyber-attack eviction in under $550\,\text{ms}$, and eliminates tire skid on low-friction road surfaces. Future work will focus on full cross-domain air-ground collaborative teams, deploying aerial scouts to provide beyond-line-of-sight safety preview for autonomous ground fleets.

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